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Literature Review on

"A Review on Deep Learning Approaches for EEG-Based Emotion Recognition"

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Supervisor Recommendation

This is to certify that Mr. Mahesh Singh Madai (Symbol no. 8115021) has submitted the literature review report on the topic **"A Review on Deep Learning Approaches for EEG-Based Emotion Recognition"** for the partial fulfilment of Master of Science in Computer Science and Information Technology, third semester. I hereby, declare that this seminar report has been approved.

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Letter of Approval

This is to certify that the literature review report prepared by Mr. Mahesh Singh Madai entitled **"A Review on Deep Learning Approaches for EEG-Based Emotion Recognition"** in partial fulfilment of the requirements for the degree of Master of Science in Computer Science and Information Technology has been well studied. In our opinion, it is satisfactory in the scope and quality as a project for the required degree.

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Mahesh Singh Madai (Exam Roll No.: 8115021)

Abstract

Emotion recognition using Electroencephalography (EEG) is a rapidly growing sub-field of affective computing, with applications in mental-health monitoring, human-computer interaction, and adaptive learning systems. Because EEG directly measures brain electrical activity, it is difficult to consciously mask, making it an attractive modality for objective emotion detection compared to facial expression or speech-based methods. However, EEG-based emotion recognition remains challenging due to high inter-subject and inter-session variability, low signal-to-noise ratio, and the absence of a standardized evaluation protocol across studies. This review systematically examines deep learning approaches for EEG-based emotion recognition, spanning an explainability-driven diagnostic study comparing classical ML, CNN, and EEGNet under SHAP analysis, and domain-adaptive approaches designed explicitly for cross-subject generalization. Comparative analysis of eight representative studies on the SEED and DEAP benchmark datasets shows that reported accuracies vary widely (from the low 60s to the upper 90s under subject-dependent evaluation) largely as a function of validation protocol rather than architecture alone, with subject-independent evaluations — including an 84% CNN result explained via SHAP feature-variance analysis — consistently yielding lower, but more trustworthy, performance than subject-dependent ones. Despite substantial architectural progress, this review identifies persistent gaps in standardized cross-study comparison, subject-independent validation, explainability, and portable low-channel deployment. This review provides a consolidated understanding of current research trends and lays the methodological groundwork for a subsequent comparative benchmarking study.

Keywords: *EEG, emotion recognition, affective computing, deep learning, graph convolutional network, transformer, subject-independent generalization.*

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LIST OF ABBREVIATIONS

BCI	Brain-Computer Interface
BiLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
DE	Differential Entropy
DEAP	Database for Emotion Analysis using Physiological Signals
DGCNN	Dynamical Graph Convolutional Neural Network
EEG	Electroencephalography / Electroencephalogram
EMG	Electromyogram
EOG	Electrooculogram
GCN	Graph Convolutional Network
ICA	Independent Component Analysis
LOSO	Leave-One-Subject-Out
LSTM	Long Short-Term Memory
ML	Machine Learning
PSD	Power Spectral Density
SEED	SJTU Emotion EEG Dataset
SVM	Support Vector Machine

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CHAPTER 1: INTRODUCTION

1.1 Introduction

Emotion recognition is the task of automatically identifying a person's affective state — typically framed along dimensions such as valence (positive/negative) and arousal (calm/excited), or as discrete categories such as happy, sad, neutral, and fear. As affective computing has matured, emotion recognition has moved beyond a purely academic pursuit into systems people already interact with. In mental-health care, wearable emotion-monitoring tools are being explored to flag early signs of depression or anxiety between clinical visits, giving therapists continuous insight rather than relying solely on infrequent self-reported check-ins. In adaptive e-learning platforms, real-time detection of a student's frustration or boredom allows the system to adjust content difficulty or pacing before disengagement leads to dropout — an increasingly sought-after capability in online education. In the automotive industry, driver-state monitoring systems use affective signals to detect drowsiness, stress, or road rage and intervene before these states contribute to accidents, a use case now being integrated into next-generation vehicle safety standards. Emotion-aware human-computer interaction also underpins more responsive virtual assistants, adaptive difficulty in gaming, and assistive communication tools for individuals with limited capacity to express emotion outwardly, such as those with autism spectrum disorder or severe motor impairment.

While emotion can be inferred from facial expressions, speech, and text — channels already exploited in many of the applications above — these signals can be consciously controlled or masked, which limits their reliability in exactly the contexts where accuracy matters most: clinical assessment, deception-sensitive settings, and safety-critical monitoring where a person's genuine internal state matters more than their outward presentation. Electroencephalography (EEG), by contrast, measures electrical activity generated directly by the brain through electrodes placed on the scalp, making it comparatively difficult to fake and therefore a valuable, objective signal for affective state estimation in precisely these high-stakes, real-world settings. EEG-based emotion recognition typically follows a pipeline of signal acquisition, preprocessing (artifact removal, filtering), feature extraction (spectral, spatial, or learned features), model training, and classification. Early work relied on handcrafted spectral features (power spectral density, differential entropy) combined with classical machine learning classifiers such as Support Vector Machines. In recent years, deep learning methods —

convolutional neural networks, recurrent networks, graph neural networks, and Transformer-based architectures — have progressively replaced handcrafted pipelines, learning discriminative spatial, spectral, and temporal patterns directly from raw or minimally processed EEG.

This review focuses specifically on EEG-based emotion recognition using the two most widely adopted public benchmark datasets, SEED (SJTU Emotion EEG Dataset) and DEAP (Database for Emotion Analysis using Physiological Signals), and comparatively analyzes eight representative deep learning approaches spanning compact convolutional networks, graph-based models, CNN-Transformer hybrids, explainability-driven diagnostic frameworks, and domain-adaptive methods, with particular attention to how validation protocol (subject-dependent versus subject-independent) affects reported performance.

1.2 Problem Statement

EEG-based emotion recognition faces several unresolved challenges. First, EEG signals exhibit substantial inter-subject and inter-session variability — the same emotional state can produce measurably different EEG patterns across individuals and across recording sessions for the same individual, making models trained on one set of subjects generalize poorly to new, unseen subjects. Second, published results across the literature are difficult to compare directly, since studies differ in preprocessing pipelines, feature representations, train/test splitting strategy, and — critically — whether evaluation is subject-dependent (train and test segments from the same subjects) or subject-independent (test subjects entirely withheld during training). This inconsistency means that headline accuracy figures reported by different papers, sometimes approaching 100%, are not necessarily comparable or indicative of real-world, cross-subject performance. Third, most architectures reporting state-of-the-art accuracy are evaluated only on the SEED and DEAP datasets under laboratory conditions using high-density electrode montages (32–62 channels), leaving open questions about performance on portable, low-channel consumer-grade EEG devices that are more realistic for real-world deployment. There is a need to systematically review these approaches, make explicit the validation protocol under which each result was obtained, and identify which architectural directions are best suited for genuinely generalizable, deployable EEG emotion recognition.

1.3 Objective

1. To review and comparatively analyze deep learning approaches — including compact CNNs (EEGNet), graph-based models (DGCNN, Siam-GCAN), CNN-Transformer hybrids, explainability-driven diagnostic frameworks, and domain-adaptive methods — for EEG-based emotion recognition.
2. To identify and evaluate the validation protocols (subject-dependent versus subject-independent) used across studies and their effect on reported performance.

1.4 Report Organization

This report is structured into four chapters. Chapter 2 reviews background concepts and related literature on EEG signal fundamentals and deep learning approaches for EEG-based emotion recognition. Chapter 3 outlines the methodology, including the literature selection strategy, inclusion and exclusion criteria, and the data extraction and synthesis process. Chapter 4 concludes with a summary of findings and identifies major research gaps across the reviewed literature.

CHAPTER 2: BACKGROUND STUDY AND LITERATURE REVIEW

2.1 Background Study

2.1.1 Emotion Recognition and Affective Computing

Affective computing is the interdisciplinary field concerned with systems that can recognize, interpret, and respond to human emotion. Physiological-signal-based emotion recognition (EEG, ECG, GSR, EMG) is generally considered more objective than expression-based recognition (facial, speech, text) because physiological signals are involuntary and difficult to consciously control. Among physiological modalities, EEG is the most direct measure of neural activity underlying emotional processing, which motivates its widespread use despite the practical challenges of signal acquisition and noise.

Most EEG-based emotion recognition systems are grounded in Russell's circumplex model of affect, which represents emotional states as points in a continuous two-dimensional space defined by valence and arousal, allowing discrete emotions such as happiness or fear to be mapped onto a shared coordinate system rather than treated as entirely separate categories. This dimensional framing aligns naturally with observed EEG correlates of emotion — most notably frontal alpha asymmetry, where relatively greater left-frontal activity is associated with approach-oriented, positive-valence states and greater right-frontal activity with withdrawal-oriented, negative-valence states, a relationship first proposed by Davidson and subsequently used as a theoretical basis for many feature-engineering choices in the datasets and models reviewed in this study. However, the mapping between neural activity and subjective emotional experience is neither perfectly linear nor fully understood, and individual differences in baseline brain activity, skull thickness, and electrode placement introduce substantial variability — a factor that, as later sections of this review will show, contributes directly to the cross-subject generalization challenges that recur throughout the EEG emotion recognition literature.

2.1.2 EEG Signal Fundamentals

EEG signals are typically decomposed into frequency bands — delta (0.5–4 Hz), theta (4–8 Hz), alpha (8–13 Hz), beta (13–30 Hz), and gamma (30+ Hz) — each associated with different cognitive and

affective states. Feature extraction for emotion recognition commonly uses Power Spectral Density (PSD) and Differential Entropy (DE) computed per band and per channel. Multi-channel montages (32 channels for DEAP, 62 channels for SEED) allow spatial features to be extracted across the scalp, which graph-based and CNN-based architectures exploit by modeling inter-channel relationships explicitly.

EEG is recorded using small scalp electrodes, most commonly arranged according to the International 10-20 system, a standardized placement scheme in which electrode positions are defined as 10% or 20% intervals of measured distances between anatomical landmarks on the skull, ensuring consistent, comparable electrode placement across subjects and studies [9]. During acquisition, electrodes detect voltage differences on the order of 10–100 microvolts, which are amplified, band-pass filtered to remove noise and drift, and digitized at a sampling rate typically between 128 and 512 Hz depending on the recording system — for example, SEED uses a 1000 Hz recording rate downsampled to 200 Hz, while DREAMER's consumer-grade Emotiv EPOC records at 128 Hz [9]. This standardization of electrode placement and acquisition parameters is what allows the datasets and architectures reviewed in this study to be meaningfully compared despite being collected by different research groups using different hardware.

The EEG acquisition process typically follows four stages [9]:

- **Preparation:** The scalp is cleaned and lightly abraded to reduce impedance, and conductive gel or paste is applied to ensure good electrical contact between the electrode and skin.
- **Electrode Placement:** Electrodes are positioned according to the 10-20 system — often via a pre-configured elastic cap for efficiency and consistency — and connected via wires to the EEG amplifier.
- **Signal Acquisition:** Electrodes detect tiny voltage differences through differential amplification, comparing each active electrode against a reference. The resulting signal is amplified (typically on the order of several thousand times), band-pass filtered (commonly around 0.5–70 Hz) to remove drift and high-frequency noise, digitized at a sampling rate typically between 128 and 512 Hz, and recorded to a computer for further processing.
- **Montages:** Post-recording, signals can be reviewed in different montage configurations — bipolar (comparing adjacent electrode pairs) or referential (each electrode compared against a common reference) — depending on the analysis or interpretation required.

2.2 Deep Learning for EEG-based Emotion Recognition

A typical deep learning pipeline for EEG emotion recognition proceeds through the following stages:

- **EEG Acquisition:** Multi-channel EEG recorded while participants view emotion-eliciting stimuli (video clips, images), labeled by self-reported or pre-validated valence/arousal/discrete-emotion ratings.
- **Preprocessing:** Band-pass filtering, and artifact removal (commonly Independent Component Analysis, ICA) to remove ocular (EOG) and muscular (EMG) contamination.
- **Feature Extraction:** Differential entropy or PSD per band/channel for classical and many CNN pipelines; raw or minimally processed multi-channel signal segments for end-to-end deep models.
- **Model Architecture**:
    - (a) Compact CNNs such as EEGNet, which factorize temporal and spatial filtering into depthwise and separable convolutions for parameter efficiency;
    - (b) Graph-based models such as DGCNN and Siam-GCAN, which represent electrodes as graph nodes and learn or exploit inter-channel connectivity;
    - (c) CNN-Transformer hybrids, which combine local convolutional feature extraction with global self-attention;
    - (d) Explainability-driven diagnostic frameworks, which pair classical ML and CNN baselines with post-hoc explanation methods such as SHAP to analyze why subject-independent performance degrades;
    - (e) Domain-adaptive models, which explicitly minimize the distributional gap between subjects to improve subject-independent generalization.
- **Classification Output:** Discrete emotion categories (e.g., positive/neutral/negative) or valence-arousal-dominance regression/classification, depending on dataset labeling.
- **Evaluation:** Accuracy, Precision, Recall, and F1-Score, computed under either subject-dependent (k-fold within-subject) or subject-independent (leave-one-subject-out or fixed train/test subject split) protocols.

Section 2.2 reviews eight representative studies spanning this architectural landscape, published

between 2018 and 2026, and Table 2.1 summarizes them for direct comparison.

2.2. Literature Review

The study [1] by Song et al. introduced the Dynamical Graph Convolutional Neural Network (DGCNN) for EEG emotion recognition, representing multichannel EEG as a graph over electrodes and learning the adjacency matrix — rather than fixing it to physical electrode distance — during network training. Evaluated on the SEED and DREAMER datasets using differential entropy features, DGCNN achieved 90.4% average recognition accuracy on SEED under a subject-dependent protocol, outperforming SVM and static-graph GCN baselines. The study established that dynamically learned inter-channel relationships capture more discriminative structure than fixed geometric graphs, and the architecture has since become a standard baseline against which newer graph-based approaches are compared. Its principal limitation is that reported results are subject-dependent, leaving open how well the learned graph transfers to unseen subjects.

In [2], Zeng et al. proposed Siam-GCAN, a Siamese (twin-branch, weight-shared) graph convolutional network with multi-head attention applied over paired EEG trials rather than isolated samples. By processing trial pairs and sharing weights across branches, the model achieved 89.23% accuracy on SEED under a subject-independent evaluation protocol — a notably strong result given that subject-independent accuracy is typically substantially lower than subject-dependent accuracy for the same dataset. This result directly targets the field's most persistent challenge: generalizing across previously unseen individuals rather than merely fitting within-subject variation.

The authors in [3] developed CIT-EmotionNet, a CNN-Transformer Interactive network designed to jointly exploit local spatial-spectral-temporal features (via CNN) and global contextual dependencies (via Transformer self-attention), with an interaction module allowing the two branches to exchange information. Evaluated on SEED and SEED-IV, the model achieved 98.57% and 92.09% accuracy respectively, outperforming prior state-of-the-art baselines including SST-EmotionNet and MDGCN-SRCNN. While these figures represent some of the highest reported accuracies in the literature, they were obtained under subject-dependent evaluation, and the paper does not report cross-subject or

leave-one-subject-out performance, limiting conclusions about real-world generalizability.

Study [4], by Margaret et al., systematically compared classical ML (KNN, Random Forest, LDA), a CNN, and EEGNet on DEAP and SEED, explicitly reporting both subject-dependent and subject-independent accuracy rather than a single headline figure. The CNN reached 98.7% under subject-dependent evaluation on SEED but dropped to a more realistic 84% under subject-independent testing, while performance on DEAP was consistently weaker under the subject-independent setting. To explain this gap, the authors applied SHAP (SHapley Additive exPlanations) and found that DEAP's SHAP-value variance across subjects was substantially higher and less consistent than SEED's, providing direct, feature-level evidence for why cross-subject generalization fails more severely on DEAP. Unlike studies reporting a single high accuracy figure, this work's transparent reporting of the subject-dependent/independent gap — paired with an explanatory mechanism — makes it one of the more methodologically trustworthy entries in this review.

In [5], Dolgopolyi and Chatzipanagiotou addressed a practical deployment gap by building a CNN-Transformer architecture that operates on only 5 of the standard 62 SEED electrodes, mapped to a consumer-grade EMOTIV Insight headset layout. Training on a merged, culturally diverse dataset combining SEED, SEED-FRA, and SEED-GER (1,455 samples total), the model achieved 90.82% test accuracy and a macro-F1 of 0.9222, outperforming SVM, DNN, and Logistic Regression baselines. This work is notable for demonstrating that competitive accuracy does not require dense, laboratory-grade electrode montages, directly supporting the feasibility of affordable, at-home EEG emotion recognition.

The authors in [6] proposed Domain Adaptive Deep Possibilistic Clustering (DADPc), reformulating Maximum Mean Discrepancy-based domain adaptation as a one-centroid, fuzzy-entropy-regularized clustering task, explicitly designed to be robust to the noise-induced domain shifts that standard MMD-based methods are sensitive to. Evaluated across SEED, SEED-IV, and DEAP under both cross-subject and cross-session protocols, DADPc showed improved recognition accuracy and generalization relative to MMD-based domain-adaptation baselines. Unlike the architecture-focused studies above, this work targets the underlying distributional mismatch between subjects directly,

representing a complementary rather than competing research direction.

Study [7], published in 2026, introduced an Enhanced Transformer-CNN-BiLSTM architecture incorporating dual attention mechanisms, combining convolutional, recurrent, and self-attention components within a single pipeline. The work reported improved performance over standalone CNN, BiLSTM, and Transformer baselines, reflecting the field's continuing trend toward hybrid, multi-family architectures. As the most recent work reviewed here, it illustrates that architectural novelty in 2026 is increasingly incremental — combining previously validated components — rather than introducing fundamentally new mechanisms.

Finally, [8] revisits EEGNet, Lawhern et al.'s compact convolutional neural network originally designed for general-purpose Brain-Computer Interface classification. By factorizing temporal and spatial filtering into depthwise and separable convolutions, EEGNet achieves competitive cross-paradigm accuracy with roughly 1,000 parameters — several orders of magnitude fewer than the Transformer- and hybrid-based models reviewed above. Although not purpose-built for emotion recognition, EEGNet is frequently used as a lightweight baseline in emotion-recognition comparisons and represents the efficiency end of the accuracy-versus-complexity trade-off that this review's synthesis (Table 2.1) makes explicit.

Table 2.1 Literature Review Comparison

Cite.	Dataset	Method / Model	Compared With	Performance	Key Finding
[1]	SEED / DREAMER; multichannel EEG (62-ch), differential entropy features	Dynamical Graph Convolutional Neural Network (DGCNN) — learns a channel adjacency matrix during training instead of using fixed electrode distance	SVM, DBN, static-graph GCN	90.4% (SEED, subject-dependent)	Learning the graph structure end-to-end outperformed fixed-distance graphs, establishing graph-based spatial modeling as a strong direction for EEG emotion recognition.
[2]	SEED; 62-channel EEG, trial-pair	Siam-GCAN — Siamese (twin-branch, weight-shared) GCN with multi-	DGCNN, RGNN, standard GCN	89.23% (SEED, subject-independent)	Reported one of the strongest subject-independent results in the field by pairing trials rather than

Cite.	Dataset	Method / Model	Compared With	Performance	Key Finding
	sampling	head attention over the electrode adjacency graph			training on isolated samples, directly addressing cross-subject generalization.
[3]	SEED / SEED-IV; differential entropy features	CIT-EmotionNet — CNN-Transformer Interactive network fusing local (CNN) and global (Transformer) spatial-spectral-temporal features	SST-EmotionNet, EeT, MDGCN-SRCNN	98.57% (SEED) / 92.09% (SEED-IV)	Combining local CNN features with global Transformer attention outperformed either family alone, though results were reported under subject-dependent, within-dataset evaluation.
[4]	DEAP (32-ch) and SEED (62-ch, reduced to 14-ch for cross-dataset testing); PSD features across five EEG bands	Classical ML (KNN, Random Forest, LDA), CNN, and EEGNet, with SHAP-based explainability used to diagnose the subject-dependent vs. subject-independent performance gap	KNN, Random Forest, LDA, EEGNet	98.7% (SEED, subject-dependent) vs. 84% (SEED, subject-independent); suboptimal on DEAP subject-independent	Directly diagnoses, via SHAP variance analysis, why subject-independent accuracy drops — showing DEAP's substantially higher inter-subject SHAP-value variance (vs. SEED's more stable variance) explains its poorer generalization, offering evidence-based insight rather than an unexplained accuracy figure.
[5]	SEED + SEED-FRA + SEED-GER merged (1,455 samples); reduced to 5 of 62 electrodes	CNN-Transformer architecture mapped to a 5-electrode consumer-grade layout (EMOTIV Insight)	SVM, DNN, Logistic Regression	90.82% test accuracy, macro-F1 0.9222	Demonstrated that competitive accuracy is achievable with a drastically reduced, consumer-grade electrode set and a culturally-merged multi-country dataset, directly relevant to low-cost, real-world deployment.
[6]	SEED, SEED-IV, DEAP; cross-subject and cross-	Domain Adaptive Deep Possibilistic Clustering (DADPc) — reframes Maximum Mean	MMD-based domain adaptation, transfer-	Improved accuracy and generalization over MMD	Directly targets the field's core weakness — inter-subject EEG variability — rather than optimizing within-subject

Cite.	Dataset	Method / Model	Compared With	Performance	Key Finding
	session protocols	Discrepancy domain adaptation as a fuzzy-entropy-regularized clustering task	component analysis baselines	baselines in cross-subject/cross-session settings	accuracy, offering a domain-adaptation-first alternative to architecture-first approaches.
[7]	Public multi-dataset benchmark (task not restricted to a single EEG corpus)	Enhanced Transformer-CNN-BiLSTM architecture with dual attention mechanisms	Standalone CNN, BiLSTM, and Transformer baselines	Reported improvement over single-architecture baselines (2026)	Represents the most recent (2026) architectural trend of combining three model families — convolutional, recurrent, and attention-based — with dual attention rather than proposing a single new mechanism.
[8]	Cross-paradigm BCI datasets (originally motor imagery / P300 / SSVEP, later adapted to other EEG paradigms)	EEGNet — compact CNN using depthwise and separable convolutions to factorize temporal and spatial filtering (~1,000 parameters)	Deeper CNNs (DeepConvNet), filter-bank CSP pipelines	Competitive accuracy across multiple BCI paradigms with orders-of-magnitude fewer parameters	Established the field's standard lightweight baseline; frequently re-used (including in emotion recognition) as the reference point that larger, more complex architectures are measured against.

CHAPTER 3: METHODOLOGY

The methodology for this literature review was designed to systematically identify, evaluate, and synthesize research on deep learning approaches for EEG-based emotion recognition, spanning compact CNNs, graph-based models, CNN-Transformer hybrids, explainability-driven diagnostic frameworks, and domain-adaptive methods. The process involved three main steps to ensure a comprehensive and reproducible analysis of the current research landscape.

3.1. Selection of Literature

A systematic search was conducted across multiple academic databases and digital libraries to identify relevant peer-reviewed literature. Primary sources included IEEE Xplore, ACM Digital Library, PubMed Central, Frontiers, Scientific Reports (Nature), arXiv, and Google Scholar. The search focused on works published from 2018 onward, capturing the field's evolution from early graph-based models (DGCNN, 2018/2020) through to the most recent Transformer- and dual-attention-based hybrids (2026).

The search strategy employed the following keywords and their combinations:

- "EEG emotion recognition deep learning"
- "EEG" AND "emotion recognition" AND "SEED" OR "DEAP"
- "Graph convolutional network" AND "EEG emotion"
- "Transformer" AND "EEG" AND "emotion recognition"
- "subject-independent" OR "cross-subject" AND "EEG emotion recognition"
- "domain adaptation" AND "EEG-based emotion recognition"

3.2. Inclusion and Exclusion Criteria

Inclusion Criteria

- Model Architecture: Studies using deep learning models (CNN, GCN, LSTM/BiLSTM, Transformer, or hybrid combinations) for EEG-based emotion recognition.
- Methodology: Research reporting a clearly stated validation protocol (subject-dependent, subject-independent, or cross-session).
- Empirical Evidence: Works presenting quantitative results using metrics such as Accuracy, Precision, Recall, or F1-Score.

- Publication Timeline: Works published from 2018 onward, with emphasis on recent (2023–2026) studies reflecting current state-of-the-art approaches.

Exclusion Criteria

- Traditional Approaches Only: Studies using solely classical ML (SVM, KNN) without any deep learning component.
- No Experimental Validation: Purely conceptual discussions, position papers, or studies without quantitative results on a public benchmark dataset.
- Limited Relevance/Detail: Duplicates, short abstracts, or insufficiently detailed studies where methodology or dataset could not be verified.
- Other Modalities: Research focused solely on other physiological signals (ECG, GSR, EMG) without an EEG component, or on non-affective EEG tasks (e.g., motor imagery only).

3.3. Data Extraction and Synthesis

Data from each selected study were systematically extracted to facilitate comparative analysis and synthesis. The key information included:

- Dataset(s) used, including source, size, channel count, and labeling scheme (e.g., SEED's 62-channel 3-class labels vs. DEAP's 32-channel valence-arousal-dominance labels).
- Preprocessing and Feature Extraction techniques specific to EEG emotion recognition, including artifact removal (ICA), band-power/differential-entropy feature extraction, and electrode-count reduction.
- Model architecture family (compact CNN, graph-based, CNN-Transformer hybrid, explainability-driven, or domain-adaptive) and key architectural innovation.
- Validation protocol — explicitly recorded as subject-dependent, subject-independent, or cross-session, since this was found to materially affect reported performance.
- Performance metrics reported (Accuracy, Precision, Recall, F1-Score).

The synthesized data, presented in Table 2.1, enabled identification of trends across architecture families and, critically, exposed the inconsistency in validation protocol across studies — a central finding that motivates the gap statement in Chapter 4

CHAPTER 4: CONCLUSION

The literature reviewed shows a clear architectural progression in EEG-based emotion recognition: from graph-based spatial modeling (DGCNN, Siam-GCAN) through CNN-Transformer hybrids and explainability-driven diagnostic studies to, most recently, multi-family dual-attention architectures. Reported accuracies on SEED and DEAP range widely, from about 62% for a linear SVM baseline up to 98.7% for a subject-dependent CNN — but [4]'s own subject-independent test of the same architecture drops to 84%, directly illustrating this review's central finding: this spread is driven substantially by validation protocol rather than architecture alone. Studies reporting near-perfect accuracy consistently rely on subject-dependent evaluation without leave-one-subject-out or cross-subject validation, while studies that do report subject-independent results (Siam-GCAN at 89.23%; the CNN in [4] at 84%; domain-adaptive methods such as DADPc) show measurably lower, but more trustworthy, performance, mirroring a broader pattern in EEG-based deep learning where aggregate accuracy can mask hidden overfitting.

Major gaps identified include: the absence of a standardized cross-study comparison protocol; inconsistent or missing subject-independent validation among the highest-accuracy studies; limited exploration of reduced-channel, consumer-grade deployment; and minimal attention to explainability outside graph-based models and SHAP-driven diagnostic studies such as [4]. Future research — including the subsequent thesis this review is intended to support — should prioritize a standardized benchmarking study under a single subject-independent protocol, reporting accuracy alongside parameter count and computational cost, and evaluating reduced-channel electrode configurations for real-world deployability.

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APPENDIX

Table 2 Experimental Protocol and Reporting Comparison Across Reviewed Studies

Study	Dataset	# Subjects	# Trials/Session	Sampling Freq.	Preprocessing	Input Representation	Train/Test Split	# Classes	Class Distribution
[1] DGCNN	SEED, DREAMER	15 (SEED)	15/session × 3 sessions	1000 Hz → 200 Hz	Band-pass 0–75 Hz, DE features	DE features (5 bands × 62 ch)	Subject-dependent, within-session	3 (SEED)	Balanced (SEED design)
[2] Siam-GCAN	SEED	15	15/session	200 Hz (standard SEED)	DE features	DE features	Subject-independent (protocol not fully specified)	3	Balanced
[3] CIT-EmotionNet	SEED, SEED-IV	15 (SEED) / 15 (SEED-IV)	15 (SEED) / 24 (SEED-IV)	200 Hz (standard SEED)	DE features	DE features	Subject-dependent	3 (SEED) / 4 (SEED-IV)	Balanced/near-balanced
[4] Margaret et al.	SEED, DEAP	Full SEED/DEAP cohorts	Standard dataset trials	200 Hz (SEED) / 128 Hz (DEAP)	ICA + band-pass, DE/PSD	DE/PSD + raw (multi-model)	Both SD and SI explicitly compared	3 (SEED) / binary (DEAP)	Balanced (SEED)
[5] Dolgopolyi et al.	SEED+FRA+GER (merged)	Merged multi-country cohort	Standard SEED trials	200 Hz (SEED-derived)	Not fully detailed	Raw signal, 5-electrode subset	Held-out test split	3	Balanced
[6] DADPc	SEED, SEED-IV, DEAP	Full cohorts per	Standard per dataset	Dataset-standard	MMD-based domain	DE features	Cross-subject/cro	3/4/binary per	Balanced per dataset

Study	Dataset	# Subjects	# Trials/Session	Sampling Freq.	Preprocessing	Input Representation	Train/Test Split	# Classes	Class Distribution
		dataset			alignment		ss-session	dataset	
[7] Karim et al.	Multi-dataset (unspecified)	NR	NR	NR	NR	NR	NR	NR	NR
[8] EEGNet	Cross-paradigm BCI	Varies by paradigm	NR	Paradigm-dependent	Standard BCI preprocessing	Raw signal	NR	Paradigm-dependent	NR